APEX ANALYTICA
MANIFOLD

CAUSAL-GRAPH FRAGILITY ANALYTICS

Manifold for Insurance & Reinsurance

A causal-inference research terminal for catastrophe modeling, treaty design, and cascade analysis across the reinsurance tower.

CONFIDENTIAL
PREPARED FOR **AON**
2026-05-08

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THE GAP

Why current CAT and risk tools fall short under regime change.

- **CAT models are statistical, fitted on historical event sets.** They break when climate non-stationarity invalidates the i.i.d. assumption underpinning every fit.
- **Cross-peril correlations are inferred from data-thin substrates.** They degrade exactly when the system is stressed — when coupling matters most, the calibration fails first.
- **Sanctions and treaty cascade are not represented as graph topology.** Today they're shoehorned into scenario libraries — additive bumps to an exposure number that doesn't know the structure exists.
- **VaR and stress tests answer *how big*. They don't answer *through which channels*.** When a primary insurer fails, every downstream treaty doesn't fail at the same severity. The structure determines the cascade — and the structure isn't in the model.

Stress tests answer how big a shock could be. Causal graphs answer through which channels it propagates.

Both are necessary. Most desks have only the first. Manifold is the structural counterpart that maps the network the rest of the stack assumes away — built by a team that comes from inside the financial-modelling discipline, not adjacent to it.

THE PLATFORM

Four engines, one configurable platform.

SPIRTES · CAUSAL DISCOVERY

Infers causal structure from claim, exposure, and balance-sheet panels. PCMCI on lagged returns and exposure deltas surfaces the dependencies the historical fit missed.

TARSKI · FORMAL VERIFICATION

Verifies treaty terms, regulatory regimes, and ILS / parametric trigger structures against the inferred graph. Red-lines connections that violate stated constraints.

PEARL · COUNTERFACTUAL INTERVENTION

Do-calculus on the graph. Answers “if I cut this exposure, what does the cascade look like?” — the question stress tests gesture at and never actually answer.

PARETO · CASCADE SIMULATION

Per-node criticality estimators (BOCPD, persistent homology, transfer entropy) plus full cascade simulation across the tower under stress.

THE SCORING FRAMEWORK

ΩF – five pillars, per-node fragility.

Every node in the graph — counterparty, peril, treaty, jurisdiction, line of business — gets a score on five orthogonal pillars. The composite **ΩF** runs with bootstrap confidence intervals, so the readout shows uncertainty alongside the point estimate.

I
IRREPLACEABILITY

Substitutability of a counterparty, facility, or treaty under a freeze. Common-collateral choke points.

R
RESTORATION LATENCY

Time to replace capacity after catastrophic failure. Recapitalisation speed under stress.

J
JURISDICTIONAL HAZARD

Sanctions cascade, regulatory regime shifts, currency-zone fragmentation, cross-border CCP fragility.

C
CASCADE LOAD

Downstream exposure depth through the reinsurance tower. Common-collateral fire-sale loops.

T
TAIL DEPTH

Distributional depth beyond VaR (99.9% / EVT regime). Correlation-collapse asymmetry.

WHERE IT LANDS AT AON

Five concrete use cases across reinsurance and capital advisory.

- **CAT model augmentation.** Surface structural correlations the statistical AIR / RMS / KCC fit missed — common-peril infrastructure dependencies, climate-driven shifts in coupling, sub-portfolio cascade depth.
- **Treaty cascade analysis.** Pearl do-calculus answers “what happens to the tower if a primary insurer or a key reinsurance partner fails?” — quantified cascade with per-node fragility readouts.
- **Climate non-stationarity stress tests.** Explicit causal model of climate-loss linkages, with regime-shift detection (BOCPD) on the time series. Defensible regulatory-disclosure narrative for TCFD / NAIC / Solvency II.
- **ILS & parametric trigger structuring.** Tarski formally verifies trigger conditions against the graph. Quantifies basis risk with structural — not statistical — causality.
- **Counterparty cascade across reinsurance partners.** ΩF readout per partner ranks fragility under stress; interdiction analysis quantifies the marginal benefit of hardening any single link.

WHAT'S LIVE TODAY

Real platform. Real data. Real validation.

Manifold is deployed in production. The numbers below are from real cohort calibration, not slideware. Honest scope: the validation substrate is currently CGM and macroeconomic time-series; the insurance-domain configuration is the next deployment, scoped to AON's data shape on engagement.

0.679

AUROC · BOCPD ON D1NAMO

Bayesian online change-point detection vs. real hypoglycemia events across 9 subjects, 8,309 paired observations. Heavy miscalibration in top decile, surfaced honestly in the readout.

0.589

AUROC · F SUB-SCORE CALIBRATION

CSD-fit relevance scoring on real cohort, mean bootstrap CI half-width 0.075. Empirical sanity-check on the ± band shipped to the UI.

4 / 4

ENGINES DEPLOYED

Spirtes, Tarski, Pearl, Pareto all live in production. Insurance-domain configuration in flight; pre-production validation against AON-shaped data is part of the proposed pilot.

WHY APEX

A team built inside the discipline, not adjacent to it.

CEO & PRINCIPAL SCIENTIST

Junaid Ghauri

D.Eng., Johns Hopkins. Bayesian models, COVID-19 analytics, reinsurance risk. Ex-CTO MARK LABS, ex-GP Pareto Technologies. Authored the dissertation underpinning **Ω**F.

HEAD OF RESEARCH

Georgios Korpas

Ph.D. Mathematics, Trinity College Dublin (visiting Stanford). Ex-HSBC AI / Optimization. Quantum algorithms for collateral and continuous optimization.

AI RESEARCH SCIENTIST

Jeremy Kulcsar

Ex-HSBC AI Research. Causality, portfolio optimization, deep learning. Author of the **Ω**F pillar framework. Ecole Centrale Paris MSc CS.

SOFTWARE / RESEARCH

Daniel Mastropietro

20+ years across energy, telecom, finance, government, manufacturing. RL, optimization, model interpretability. Doctoral candidate INP Toulouse, Fulbright alumnus.

RISK MODELING ADVISOR

Dr. Gonzalo L. Pita

Associate Research Scientist, JHU. Natural disaster risk, climate change, infrastructure vulnerability. Former Senior Consultant, World Bank.

SYSTEMS ENGINEERING ADVISOR

Dr. Takeru Igusa

Professor, Johns Hopkins. Systems science, applied math, epidemiology, resilience. NIH and CDC funded across 25+ years of risk-modeling research.

HOW TO START

Four engagement shapes, sized to the question.

ΩF MINI-AUDIT

$1,500

Flat · 5 business days

Send a sanitised AON dataset. Get a 12-page **ΩF** readout, top-3 fragility nodes, suggested interdictions, and a 30-min walkthrough.

CAUSAL RISK ASSESSMENT

$7.5-15k

2–3 weeks · scoped

Custom analysis on a specific question — a treaty cascade, a peril cluster, a counterparty network. Configured Manifold instance + written deliverable.

ENTERPRISE PILOT

$150k+

3–6 months · embedded

Insurance-domain configuration of Manifold against AON's data shape. On-demand engineering, dedicated solutions support, contractual SLA.

PLATFORM LICENSE

From $48k

Per seat / year

Continuous Manifold access for an AON team — multi-domain seats with cross-asset bridges, live feeds, and the full estimator suite.

WHAT WE'RE ASKING

A 30-minute walkthrough – and one of three follow-on paths.

Show us a domain you care about. We'll bring Manifold live, configured for that domain, and walk through one cascade scenario relevant to AON's book or capital-advisory work.

- **Path 1 — ΩF Mini-Audit on a sanitised AON dataset.** $1,500 flat. Concrete output in 5 business days.
- **Path 2 — Scoped Causal Risk Assessment.** Pick one question; we deliver a written analysis in 2–3 weeks at $7.5–15k.
- **Path 3 — Enterprise Pilot.** 3–6 month embedded engagement, configured Manifold instance, $150k+. The path forward if the demo earns it.

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